

Liver Radiodensity and Time to Dose-Limiting Toxicity

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- Evaluate association between liver HU and toxicity
- Time-to-event: cycles until dose-limiting toxicity
- Adjust for clinical factors and hospital

Conclusion: Study assesses predictive value of liver radiodensity.

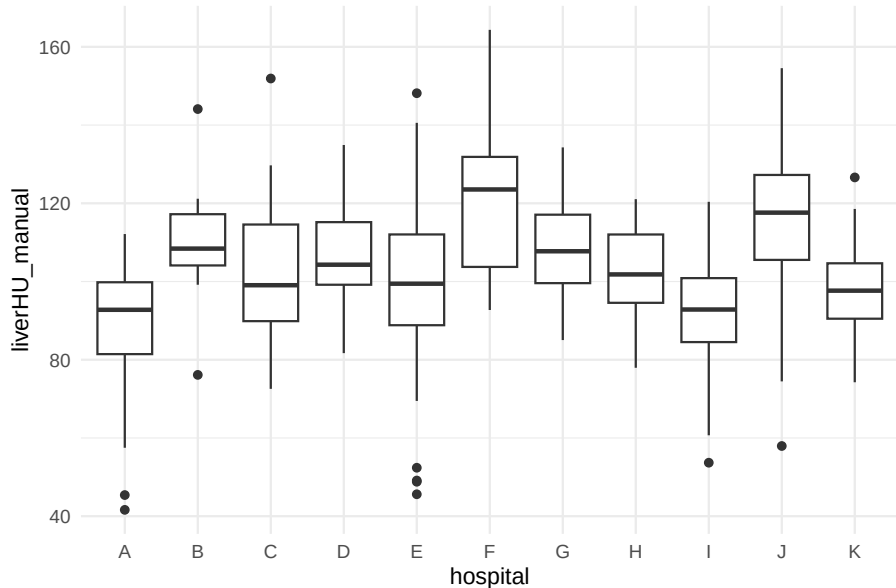
- CAP toxicity
- OXA toxicity
- CAPOX toxicity

Conclusion: Three independent survival analyses performed.

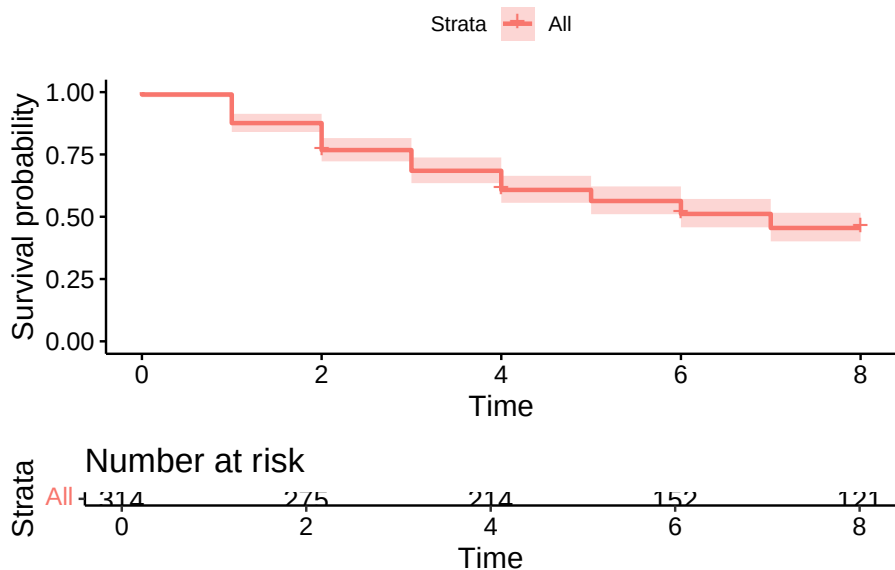
Endpoint	N	Events	Censored
CAP	344	181	163
OXA	311	254	57
CAPOX	359	257	102

Conclusion: Toxicity frequency differs across treatments.

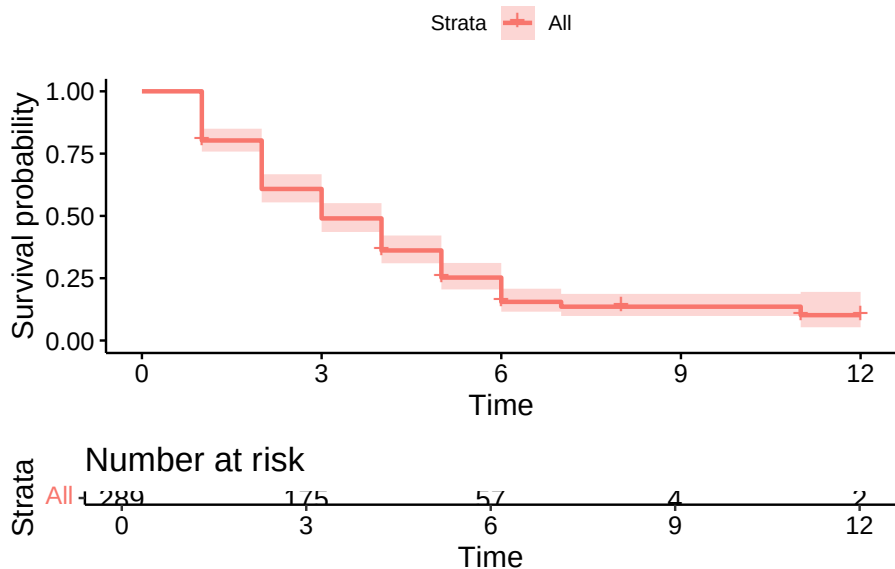
Liver HU by hospital



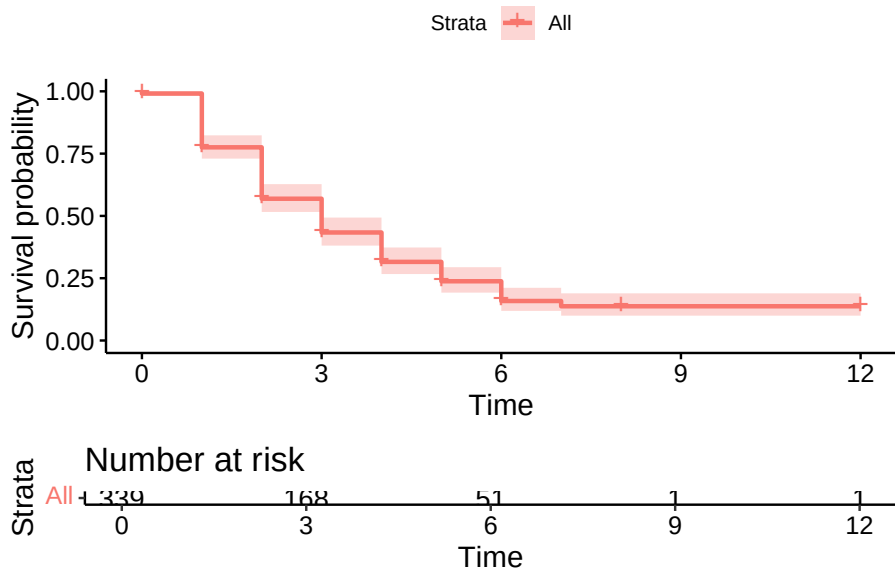
Kaplan-Meier CAP



Kaplan-Meier OXA



Kaplan-Meier CAPOX



Conclusion KM Curves

Across all treatments, most dose-limiting toxicities occur during early chemotherapy cycles, with decreasing numbers of patients at risk in later cycles leading to increased uncertainty.

Few patients remain under observation at later cycles, leading to wider uncertainty.

This justifies survival analysis

Next step

Multivariable Cox models fitted for all three endpoints.

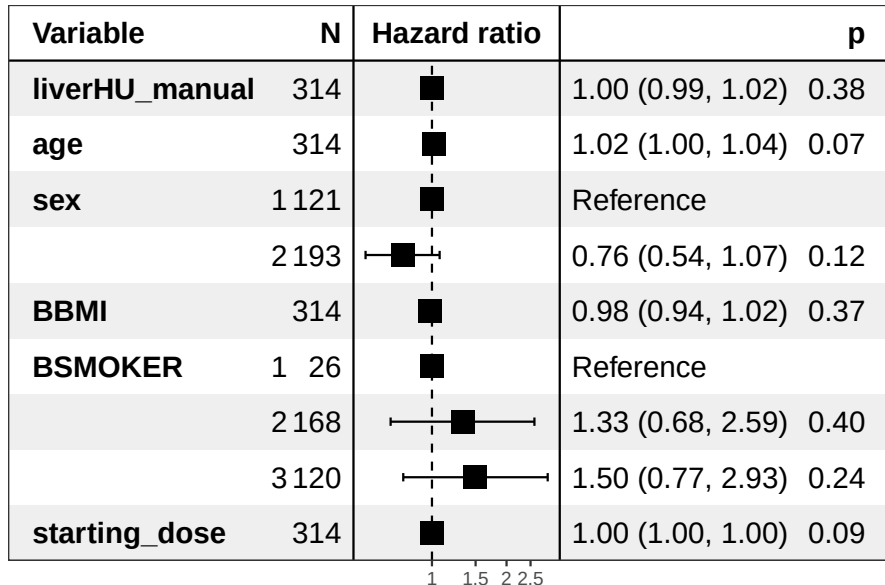
Hospital stratification allows baseline toxicity risk to differ between centers without estimating separate coefficients.

- each hospital has its own baseline hazard
- but same covariate effects

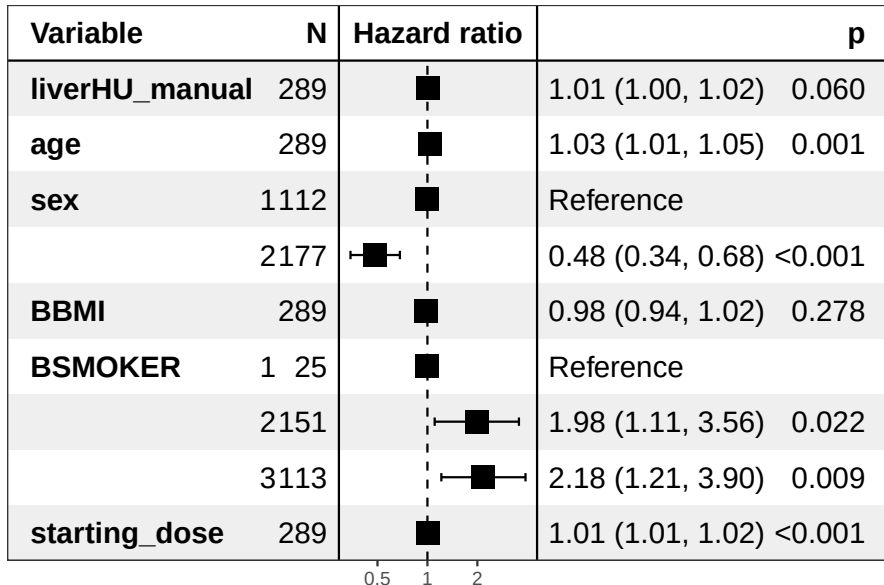
Hazard ratio table

term	estimate	std.error	statistic	p.value	conf.low	conf.high	Model
liverHU_manual	1.00	0.01	0.88	0.38	0.99	1.02	CAP
age	1.02	0.01	1.79	0.07	1.00	1.04	CAP
sex2	0.76	0.18	-1.55	0.12	0.54	1.07	CAP
BBMI	0.98	0.02	-0.89	0.37	0.94	1.02	CAP
BSMOKER2	1.33	0.34	0.83	0.40	0.68	2.59	CAP
BSMOKER3	1.50	0.34	1.18	0.24	0.77	2.93	CAP
starting_dose	1.00	0.00	1.70	0.09	1.00	1.00	CAP
liverHU_manual	1.01	0.00	1.88	0.06	1.00	1.02	OXA
age	1.03	0.01	3.18	0.00	1.01	1.05	OXA
sex2	0.48	0.17	-4.15	0.00	0.34	0.68	OXA
BBMI	0.98	0.02	-1.09	0.28	0.94	1.02	OXA
BSMOKER2	1.98	0.30	2.30	0.02	1.11	3.56	OXA
BSMOKER3	2.18	0.30	2.61	0.01	1.21	3.90	OXA
starting_dose	1.01	0.00	3.59	0.00	1.01	1.02	OXA
liverHU_manual	1.01	0.00	1.22	0.22	1.00	1.02	CAPOX
age	1.01	0.01	0.90	0.37	0.99	1.02	CAPOX
sex2	0.75	0.14	-1.99	0.05	0.57	1.00	CAPOX
BBMI	0.99	0.02	-0.59	0.55	0.96	1.02	CAPOX
BSMOKER2	1.78	0.27	2.14	0.03	1.05	3.01	CAPOX
BSMOKER3	1.83	0.27	2.24	0.03	1.08	3.12	CAPOX

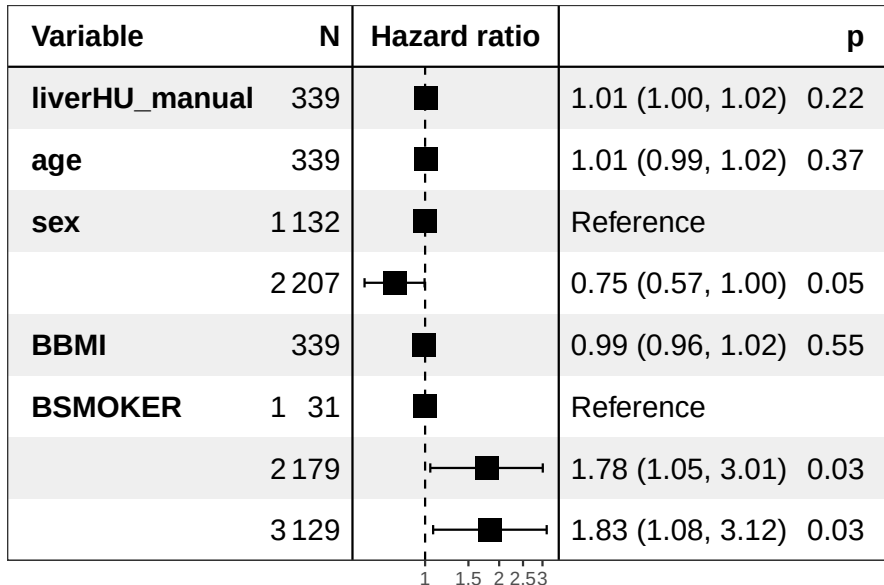
Forest plot CAP



Forest plot OXA



Forest plot CAPOX



PH assumption

The proportional hazards assumption was evaluated using Schoenfeld residual tests. Non-significant results indicate no strong evidence that hazard ratios change over time. This supports the validity of the Cox model estimates.

	chisq	df	p
liverHU_manual	0.2255962	1	0.6348086
age	0.2120542	1	0.6451621
sex	0.7255615	1	0.3943257
BBMI	0.7826862	1	0.3763209
BSMOKER	0.1230178	2	0.9403446
GLOBAL	2.0444171	6	0.9155685

Conclusion: No major PH violations.

Overall conclusion

- Liver radiodensity weakly associated with toxicity
- Effects small across all endpoints
- Dose and clinical variables stronger predictors
- Hospital differences accounted for

Final conclusion: Liver radiodensity is not a strong predictor of chemotherapy toxicity.